

**4. Chance, natural selection, and the environment interact.** Chance events can affect the subsequent evolutionary history of populations. For instance, when a storm blows insects or birds hundreds of kilometers over an ocean to an island, the wind does not necessarily transport those individuals that are best suited to the new environment. Thus, not all alleles present in the founding population's gene pool are better suited to the new environment than the alleles that are "left behind." In addition, the environment at a particular location may change unpredictably from year to year, again limiting the extent to which adaptive evolution results in organisms being well suited for current environmental conditions.

With these four constraints, evolution does not result in perfect organisms. Natural selection operates on a "better than" basis. We can, in fact, see evidence for evolution in the many imperfections of the organisms it produces.

#### CONCEPT CHECK 23.4

1. What is the relative fitness of a sterile mule? Explain.
2. Explain why natural selection is the only mechanism that consistently leads to adaptive evolution in a population.
3. **VISUAL SKILLS** Consider a population in which heterozygotes at a certain locus have an extreme phenotype (such as being larger than homozygotes) that confers a selective advantage. Compare this description to the three types of selection shown in Figure 23.13. Does this situation represent directional, disruptive, or stabilizing selection? Explain.

For suggested answers, see Appendix A.

## 23 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

### SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to [goo.gl/zk9t](http://goo.gl/zk9t).

#### CONCEPT 23.1

**Genetic variation makes evolution possible** (pp. 487–489)

- **Genetic variation** refers to genetic differences among individuals within a population.
- The nucleotide differences that provide the basis of genetic variation originate when mutation and gene duplication produce new alleles and new genes. New genetic variants are produced rapidly in organisms with short generation times. In sexually reproducing organisms, most of the genetic differences among individuals result from crossing over, the independent assortment of chromosomes, and fertilization.

? Typically, most of the nucleotide variability that occurs within a genetic locus does not affect the phenotype. Explain why.

#### CONCEPT 23.2

**The Hardy-Weinberg equation can be used to test whether a population is evolving** (pp. 489–493)

- A **population**, a localized group of organisms belonging to one species, is united by its **gene pool**, the aggregate of all the alleles in the population.
- For a population in **Hardy-Weinberg equilibrium**, the allele and genotype frequencies will remain constant if the population is large, mating is random, mutation is negligible, there is no gene flow, and there is no natural selection. For such a population, if  $p$  and  $q$  represent the frequencies of the only two possible alleles at a particular locus, then  $p^2$  is the frequency of one kind of homozygote,  $q^2$  is the frequency of the other kind of homozygote, and  $2pq$  is the frequency of the heterozygous genotype.

? Is it circular reasoning to calculate  $p$  and  $q$  from observed genotype frequencies and then use those values of  $p$  and  $q$  to test if the population is in Hardy-Weinberg equilibrium? Explain your answer.

#### CONCEPT 23.3

**Natural selection, genetic drift, and gene flow can alter allele frequencies in a population** (pp. 493–497)

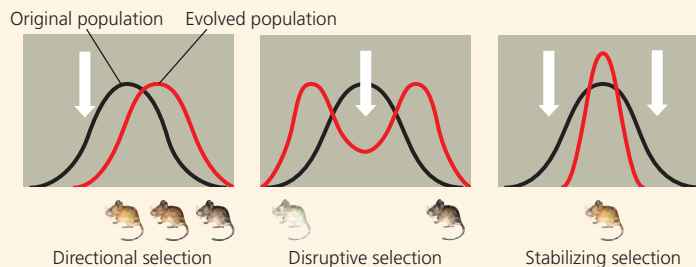
- In natural selection, individuals that have certain inherited traits tend to survive and reproduce at higher rates than other individuals *because of* those traits.
- In **genetic drift**, chance fluctuations in allele frequencies over generations tend to reduce genetic variation.
- **Gene flow**, the transfer of alleles between populations, tends to reduce genetic differences between populations over time.

? Would two small, geographically isolated populations in very different environments be likely to evolve in similar ways? Explain.

#### CONCEPT 23.4

**Natural selection is the only mechanism that consistently causes adaptive evolution** (pp. 497–504)

- One organism has greater **relative fitness** than another organism if it leaves more fertile descendants. The modes of natural selection differ in their effect on phenotype:



- Unlike genetic drift and gene flow, natural selection consistently increases the frequencies of alleles that enhance survival and reproduction, thus improving the degree to which organisms are well-suited for life in their environment.
- **Sexual selection** can result in secondary sex characteristics that can give individuals advantages in mating.
- **Balancing selection** occurs when natural selection maintains two or more forms in a population.